Grant Liu

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Education

University of California, Berkeley - B.S. Electrical Engineering & Computer Sciences May 2028

Coursework: Digital Design & Integrated-Circuit, ASIC Laboratory, Microelectronic Devices & Circuits, Signals & Systems, Integrated Circuit Devices, Data Structures, Computer Architecture, Internet Architecture, Foundations of Business, Personal Finance Management, Data & Decisions, Sustainable Business

Experiences

MiCable, *Hardware Engineering Intern* June 2025 - July 2025

- Built a **CAD-CAM-CNC** pipeline: introduced a six-sided soft-jaw fixture that cuts setup time and improves reliability.
- Stabilized RF cable measurements: performed SOLT calibrations on a VNA, verified 50-ohm terminations, logged S11/S21 baselines, and authored a torque SOP that improved run-to-run repeatability and cut retest rates.
- Improved PA bench performance: swept bias points and tuned LC input/output networks to raise small-signal gain/stability and suppress spurs; documented with Bode/Smith plots and before/after spectra for traceability.
- Hardened the test chain: mapped a 2–18 GHz signal path with dual directional couplers, built an error budget (cable loss, coupling factor, detector linearity), and flagged high-leverage tuning nodes for faster impedance matching.

UC Berkeley Computer Science Mentors, *Senior Mentor* September 2025 - Present

- Led weekly group tutoring sessions for Introduction to Circuits & Devices class to facilitate collaborative learning.
- Designed practice worksheets, host exam review sessions for students, and offer office hours for Q&A to help students.

UC Berkeley Codeology, *Project Lead* February 2025 - Present

- Authored the production requirement for a *Course Recommender* and prioritized a lean MVP ranking on quality.
- Implemented React UI and Python/Flask backend on SQLite; owned API design, schema, and auth/session handling.
- **Team leadership**: host weekly code reviews and "teach-ins"; mentored 5 members and unblocked merge-critical PRs.

Calsol UC Berkeley Solar Vehicle Club, *Electrical Team Member* September 2024 - May 2025

- Designed KiCad PCBs and owned prototype assembly/bring-up; executed precision rework and drove BOM updates.
- Debugged regenerative braking and tuned control parameters, boosting drivetrain efficiency by $\sim$**15%**.
- Iterated **5** PCBs across BMS/motor control, improving connectorization and firmware error handling to reduce faults.

Projects

Michelin Watershed Collective Action

- Benchmarked 10 geographically diverse watershed collective-action initiatives using a science-based rubric (8 criteria, 4 pillars) spanning water quantity, quality, and habitat restoration aligned to Michelin's multi-basin footprint.
- Built a scalable success framework for watershed collective action; operationalized a 12-month Mexico pilot plan, detailing stakeholder mapping, target-setting, governance, standardized monitoring, and basin-level disclosure.

Low-Cost Wide-Span Frequency Identification Method

- Replaced direct RF counting with an *insertion-loss-vs-frequency* correlation; routed the unknown through a characterized line and matched ΔP to a calibrated lookup table and validated the feasibility of **10 MHz – 110 GHz**.
- Targeted **100**$\times$ lower instrument cost than a spectrum analyzer with $\sim$**20 MHz** typical resolution; documented a full calibration package, including a test plan and step-by-step production procedure, and verified against known carriers.

Berkeley Course Recommender *- Python, HTML, CSS*

- Implemented a Python and Flask backend with SQLite schemas (courses, prerequisites, feedback); added session authentication (hashed passwords); encoded multi-factor ranking history, difficulty proxies, time conflicts.
- Built React components for search filter, result cards, and a shared-notes panel; instrumented client-side timing.

Poker Hand Win Rate Predictor *- Python, CSS*

- Wrote a Monte Carlo engine (**10k** runs configurable) modeling deck state, shuffles, and hand scoring; verified with randomized tests to guard against regression with added seed control for reproducible demonstrations.
- Produced a React and CSS UI for card entry and random board runouts with live probability updates.

Technical Skills

Software: Python, Java, C, C++, RISC-V, HTML/CSS
RF/Test: Vector Network Analyzer calibration; Electronic devices & cables testing; Torque SOPs; Impedance matching
EE/Embedded: KiCad; PCB bring-up, assembly, and rework; CAN diagnostics; Firmware debug; Soldering